DEVELOPMENT PHASE

Date	30 June 2026
Team ID	23A51A42A3
Project Name	Metro Ticket Generating System using ServiceNow
Maximum Marks	5 Marks

Introduction

The development phase is the most important stage of the Metro Ticket Generation System. During this phase, the identified requirements were converted into a functional ServiceNow application. The project was developed using various ServiceNow modules such as Service Catalog, Variables, Client Scripts, UI Pages, Jelly Scripting, Flow Designer, Custom Tables, and Database Management.

The main objective of the development phase was to provide passengers with a simple and user-friendly platform where they can either recharge their metro smart card or book a QR-based metro ticket digitally. The application eliminates manual ticket booking and provides a faster and more convenient process.

The development process was carried out in multiple stages including database creation, catalog item configuration, user interface development, workflow automation, QR code generation, testing, and validation. Every module was integrated step by step to ensure smooth communication between the user interface and backend database.

The project follows the ServiceNow development standards by separating presentation, business logic, automation, and database operations into different modules, making the application easier to maintain and extend in the future.

Development Environment

The Metro Ticket Generation System was developed using the ServiceNow cloud platform because it provides low-code application development, workflow automation, database management, and user-friendly interfaces.

The following development environment was used during implementation.

Platform:
ServiceNow Personal Developer Instance (PDI)

Development Tools:

- Service Catalog
- Flow Designer
- UI Pages
- Jelly Framework
- Client Scripts
- Catalog Client Scripts
- Custom Tables
- Variables
- Reference Fields
- Script Includes (where required)

Programming Languages:

- HTML
- CSS
- JavaScript
- Jelly XML
- Glide APIs

Database:
Custom Metro Database Table

Browser:
Google Chrome

Operating System:
Windows 11

Development Methodology:
Incremental Development

The project was developed incrementally. Each feature was implemented individually and tested before moving to the next module. This reduced errors and simplified debugging during development.

Service Catalog Development

The first module developed was the Service Catalog Item because it acts as the primary interface between passengers and the system.

A catalog item named "Book A Metro Ticket" was created inside Service Catalog.

The catalog item provides two different services.

1. Recharge Metro Card
2. Book QR Ticket

Radio button variables were used so that users can select only one service at a time.

Depending on the selected option, the corresponding input fields become visible while the remaining fields stay hidden.

For QR Ticket Booking, the following variables were created:

- Starting Station
- Destination Station
- Number of Passengers
- Journey Type
- Single Journey Fare
- Return Journey Fare

For Metro Recharge, the following variables were developed:

- Smart Card Number
- Smart Card Holder Name
- Recharge Amount
- Payment Mode

Mandatory validations were applied to prevent incomplete submissions.

Reference variables were used for station selection, allowing users to select stations from predefined values instead of entering free text.

This module provides a clean and simple interface for passengers while minimizing data entry errors.

Client Side Development

Several Catalog Client Scripts were developed to improve user experience and automate calculations.

The first client script dynamically displays the required fields depending on the selected service.

If the passenger selects "Recharge Metro Card", recharge-related variables become visible while QR booking fields remain hidden.

If "Book QR Ticket" is selected, only ticket booking fields become visible.

Another client script automatically calculates ticket fares.

The fare calculation is based on:

- Starting Station
- Destination Station
- Journey Type

The calculated amount is automatically displayed without requiring manual calculation.

Mandatory validations were also implemented.

The application checks whether:

- Starting station is selected.
- Destination station is selected.
- Passenger count is entered.
- Journey type is selected.
- Recharge amount is entered.
- Payment method is selected.

If mandatory information is missing, the system displays an appropriate validation message before allowing submission.

These client-side validations reduce incorrect entries and improve the overall user experience.

QR Code Generation Module

A custom UI Page named Metro QR Generator was developed using Jelly Framework.

The UI Page uses HTML for structure, CSS for styling, and JavaScript for dynamic QR code generation.

The open-source QRCode.js library was integrated into the project for generating QR codes.

The generated QR code contains ticket information such as:

Passenger Name

Source Station

Destination Station

Journey Type

Fare

Ticket Status

The QR code is displayed immediately after ticket generation.

The page also displays ticket information in a structured table below the QR image.

The generated ticket includes:

Passenger Name

Starting Station

Destination Station

Journey Type

Fare Amount

Ticket Status

This module demonstrates how ServiceNow UI Pages can be customized using Jelly scripting to build interactive web interfaces.

Database Development

A custom database table named Metro Database was created to store recharge information.

The following fields were created inside the table.

User Details

Smart Card Number

Smart Card Name

Recharge Amount

Mode of Payment

Created Date

Reference fields were used wherever applicable to maintain relationships between records.

The Metro Database table serves as the backend storage layer of the application.

Whenever a recharge request is successfully processed, the application stores the recharge information in this table.

This centralized database makes it easier to retrieve recharge history, perform reporting, and maintain passenger records.

Proper field types such as String, Integer, Reference, Currency, and Date/Time were selected to ensure accurate data storage.

Workflow and Flow Designer Development

Flow Designer was used to automate backend operations.

A Service Catalog Trigger was configured to execute automatically whenever a recharge request is submitted.

The workflow performs the following operations.

Receives the catalog request.

Reads submitted information.

Creates a new record inside the Metro Database table.

Stores recharge details.

Completes the process automatically without manual intervention.

Automation significantly reduces manual effort and ensures consistent record creation.

The workflow follows the event-driven architecture of ServiceNow where user actions automatically trigger backend processes.

Although the project currently includes recharge record creation, the same workflow architecture can easily be extended for ticket validation, payment confirmation, notification services, and smart card balance management.

Integration and System Development

After developing all individual modules, the entire application was integrated.

The following components were connected together.

Service Catalog

Client Scripts

Variables

Flow Designer

Metro Database

UI Pages

QR Code Generator

The integration process ensures that data flows correctly from user input to backend storage.

The overall system performs the following sequence.

The passenger opens the catalog item.

The passenger selects Recharge Metro Card or Book QR Ticket.

Relevant fields are displayed.

The user enters required information.

Client Scripts validate user input.

Fare is calculated automatically.

The request is submitted.

Flow Designer processes the request.

Recharge information is stored in the Metro Database.

For QR Ticket booking, the ticket is generated with a QR code.

This modular architecture improves maintainability and future scalability.

Development Challenges and Future Enhancements

During development, several challenges were encountered.

Initially, configuring dynamic catalog variables required careful client-side scripting.

Flow Designer variable mapping also required additional configuration to correctly retrieve catalog values.

Developing the QR code page using Jelly scripting required integrating external JavaScript libraries while maintaining compatibility with the ServiceNow platform.

Database relationships were also carefully designed to avoid duplicate records and maintain consistency.

Despite these challenges, the application was successfully developed and tested.

The current implementation provides a functional prototype that demonstrates digital metro ticket booking and smart card recharge.

Future enhancements may include:

Integration with online payment gateways.

Real-time metro station database.

Dynamic fare calculation based on distance.

Ticket expiry validation.

QR code scanning at metro gates.

SMS and email ticket notifications.

Metro card balance enquiry.

Passenger travel history.

Mobile-responsive interface.

Role-based administration dashboard.